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PRISM: A decision support system for forest planning

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ABSTRACT

PRISM, a new forest management decision support system, was developed for United States national forest planning at the same time existing tools were deprecated. PRISM has a friendly graphical user interface to facilitate model development, database management, sensitivity analysis, and data visualization. It integrates SQLITE engine for database storage and exploration. It supports both classic forest planning Model I and II to schedule management activities and achieve multiple management objectives through the lifecycle of a forest plan. Multiple models may be created through the PRISM interface and solved in batch mode for convenient sensitivity analysis. Various options to visualize model outputs are available through the embedded JFREECHART library. The open-source nature of PRISM makes it free to use and allows for long-term software development and collaboration. This paper presents the architecture, core functionality, and selected applications of PRISM in the United States national forest planning.

Software availability

Name of software: PRISM
Description: open-source decision support system for forest planning
Developers: Dung Nguyen, Eric Henderson, Yu Wei, David Anderson
Year first available: 2016
Hardware required: 64-bit computer
Software required: Java JDK 15 or later versions
Program language: Java
Program size: 72 Megabytes (version 3.2.01)
Cost and license: Free under the GNU General Public License Version 3
Availability: http://bitbucket.org/Prism-Members/prism/src/master

1. Introduction

Forest planning often includes identifying a long-term forest management direction and evaluating effectiveness of forest management practices in achieving the objectives of the management plan. Historically, forest managers have used decision support systems (DSS) to inform the development and implementation of forest management plans. In this paper, we introduce PRISM, a newly developed DSS that has been used to support planning efforts by the United States Forest Service (USFS). Here we present the architecture, core functionality, and performance of PRISM and discuss its utility in real-world applications.

The USFS administers 188 million acres of land organized into 154 national forests (Hoover, 2019). Each forest maintains a Land Management Plan (i.e., forest plan) that describes how the forest should be managed to achieve recreation, livestock grazing, timber harvesting, watershed protection, and wildlife habitat conservation objectives (USDA Forest Service, 2008). The National Forest Management Act of 1976 (16 U.S.C. 1604) requires every national forest to periodically revise its forest plan to address changing management goals, forest conditions, and public values. Plan development and revision is often carried out by an interdisciplinary planning team of natural resource experts with input from the public and stakeholders and is a complex process that may take five to seven years to complete (Haber, 2015).

A forest plan is comprised of several components, including long-term aspirational goals called desired conditions, shorter-term measurable objectives, a set of management standards and guidelines to follow, and the suitability of lands that defines where certain management activities are permissible [36 CFR 219.7,2,3]. Moving a forest from its current conditions to the desired conditions is often a complex problem that involves consideration of multiple management goals, physical and financial constraints, and the interaction of management actions and vegetation conditions through time. The best solution to this problem is not usually readily apparent.

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Evaluating multiple management strategy options and their associated effects is one way to determine a good management plan. United States national forest planning implements the National Environmental Policy Act (42 U.S.C. §§ 4321 et seq.) where alternative management strategies, such as suitability of lands or harvesting levels, are analyzed to determine their environmental, social, and economic effects. Analysis of land management plan alternatives is often supported by DSSs to provide comparative information for decision makers. By comparing the beneficial and adverse effects of different alternatives, forest managers can evaluate trade-offs and make an informed decision about which alternative should be adopted as the forest plan.

For more than half a century, DSSs have been used to assist forest managers in making decisions that account for multiple economic, ecological, social, and legal forest planning considerations (Reynolds et al., 2008; Segura et al., 2014; Packalen et al., 2013). The development and evolution of DSSs are driven by both the increasing demand for solving complex contemporary planning and the availability of advanced computer technologies (Vacik and Lexer, 2014). A 2014 synthesis of decision support systems for forest management planning reported 265 prevalent DSSs in 26 countries that can address a broad range of forest planning problems (Borges et al., 2014). Many of those rely on mathematical formulations that implement quantitative techniques such as decision analysis, simulation, and optimization (Power and Sharda, 2007; Nižetić et al., 2007).

The USFS has a long history with both simulation-based and optimization-based DSSs to support strategic planning (Borges et al., 2014; Schuster et al., 1993; Rauscher, 1999; Reynolds, 2005). In the period from 1979 to 1996 when US forest managers and policy makers emphasized systematic planning and optimal management choices, optimization-based DSSs including FORPLAN (Johnson and Stuart, 1987; Field, 1984) and its successor SPECTRUM (Greer and Meneghin, 2000) were widely used. The preference of using optimization DSSs were driven by the 1982 Planning Rule (36 CFR 219) that stipulated the use of optimization methods to determine key plan requirements such as sustainable harvest, maximum present net value, and the suitability of lands. Beginning in the late 1990s, there has been a trend to adopt simulation systems for the US national forest planning. Typical simulation-based DSSs for national forest planning in this period were VDDT and its current version SyncroSim/ST-SIM (Daniel et al., 2012), SIMPPLLE (Chew et al., 2004), and LANDIS (Hallett et al., 2017; Scheller et al., 2007; Xi et al., 2020).

Optimization models and simulation models have different strengths and associated weaknesses. While optimization models are strong at finding good strategies for achieving the desired objective, they must be told explicitly the level of disturbance to expect in the future, which cannot be exactly known in reality (Boychuk and Martell, 1996). Simulation models can be used more flexibly to reflect the uncertainty of future disturbances, but often must be given the rules for when and how much management to implement (Gustafson et al., 2000; Shang et al., 2012). Simulation models may not achieve the desired objective as efficiently as optimization models. Forest managers often have to consider these trade-offs when deciding which model to use.

In 2016, the Northern Region of the USFS and Colorado State University began collaborating to develop the optimization model PRISM as a pilot project to support the Custer Gallatin National Forest plan revision. The name “PRISM” is an acronym for “plan-level forest activity scheduling model”. This tool allows USFS managers to rapidly develop long-term forest management strategies and compare the effects of various plan alternatives. Two main factors contributed to the genesis of the PRISM project. First, there was a need to respond to the promulgation of the Forest Service’s 2012 Planning Rule (36 CFR 219) which emphasizes: 1) using the best available scientific information to inform forest planning, 2) achieving vegetation desired conditions and ecosystem integrity objectives, 3) a financial shift from maximizing present net value to completing projects within budget, and 4) completing a planning revision within a shorter length of time (USDA Forest Service, 2011; Nie, 2019). Second, the USFS planning software SPECTRUM was deprecated due to obsolete programming languages and a lack of maintenance capacity.

The resulting PRISM DSS makes several improvements over its predecessors. It incorporates better computer programming technology and contemporary decision science. Development was done with Java and an established long-term support plan. PRISM is maintained as an open-source project which provides opportunities for long-term software maintenance and developer collaboration. It incorporates enhanced modelling techniques such as a flexible Model I and II formulation structure. PRISM’s design facilitates rapid model development and streamlined debugging. The software reduces the complexity of model development by simplifying input requirements. The software user interface provides an intuitive platform to include natural disturbances in the model formulation. PRISM allows for virtually limitless attribute tracking that can be used in both the model formulation and back-end reporting. Model runs can be batched to allow for efficient sensitivity analyses and to minimize user set-up. Finally, run outputs can be readily analyzed through both predeveloped and customized graphs, charts, tables or summary queries that can be saved and used in other related model runs.

The challenges in developing PRISM were similar to those associated with developing environmental modelling DSSs, including user engagement, application of advanced technologies, product adoption, and measurement of product success (McIntosh et al., 2011). Therefore, we adopted some of the environmental DSS development best practices when developing PRISM (McIntosh et al., 2006, 2011; Laniak et al., 2013; Jakeman et al., 2006; Pastorella et al., 2015). For example, to overcome the challenge of engaging end users, we participated in several team meetings with the Custer Gallatin National Forest to ensure PRISM provided the functions required by the forest planning process. To ensure technology competitiveness, we used a contemporary programming language (Java) to build PRISM, utilized well-established methods (i.e., Model I and II in forest management (Gunn, 2007; Davis et al., 2001; Martin et al., 2017; McDill et al., 2016; Belavenutti et al., 2018; Martin, 2013)) to develop the PRISM optimization model, established connections to advanced solvers (CPLEX) for better model performance, and integrated third-party libraries to improve data management (SQLITE (Bhosale et al., 2015)) and data visualization (JFREECHART (JFreeChart Homepage, 2022)). We created an easy-to-use interface for building models and displaying model results at different stages of the planning process. This in turn creates opportunities for silviculturists, biometricians, analysts, and planners to participate in model development and validation through the entire planning process. Finally, PRISM was developed as an open-source software package so that it is inexpensive to use, update, and maintain.

This paper describes the architecture, interface design and key functionality of PRISM to emphasize the software’s utility and ease of use. Several real-world applications are discussed to showcase the software’s capabilities and performance. While initially developed for national forest planning, PRISM has the potential to support more general forest management and we expect the software will be of interest to forest owners and managers outside of the USFS. This software can be freely downloaded, utilized and customized with the acceptance of an open-source licensing agreement.

2. Forest planning analysis components and processes

Developing a PRISM model requires the user to compile two key datasets, including a description of the current state of the forest (strata) and a menu of options (yield tables) for managing forest strata through time. Forest vegetation in each of these strata is grown forward through numerous future management trajectories (prescriptions) that is compiled in “yield tables”. Information about strata and yield tables are stored as tables in a relational database in PRISM. Database information is then used to build forest planning models directly through the software interface. Users must describe any restrictions on management as the model's "constraints". A measure of solution quality is defined through the model's "objective function". The goal is then to determine the prescription(s) to choose for each stratum that stay within the constraints and result in the best objective function value possible.

The first dataset required by PRISM is a description of the forest delineated into multiple strata. Forest areas within each stratum share identical characteristics that represent a unique and meaningful ecological, economic, and/or social class. These characteristics (also called "forest attributes") can be static or dynamic depending on whether they would stay as a constant or change across time. For example, a geographic location on the forest is a static feature that represents a spatial aspect of the forest and does not change through time. Forest (tree) size class, forest species composition, and structural states are all dynamic attributes that are expected to change through time, but may be included in the strata definitions that depict the current condition of the forest.

Yield tables are the set of possible vegetation growth trajectories for each stratum resulting from different management strategies (i.e., "prescriptions"). Yield tables store dynamic attributes of the strata at each time step (or "planning period") such as timber volume, wildlife habitat suitability, or species composition. Some important dynamic attributes including vegetation type, size, and density are often used together to represent the "state" of forest strata, which in turn can be used to define the vegetation desired conditions. Each prescription in a yield table is a unique series of forest management activities applied to a stratum through time (Powell, 2013; British Columbia and Ministry of Forests, 2000; Salwasser). Management activities may include thinning, clearcut, planting, prescribed burning, herbicide and/or fertilization etc. Prescriptions must be designed such that the types and timing of management activities are technically practical and ecologically sound. For example, a harvest prescription may include a thinning activity to prepare for the final clearcut several decades later. Prescriptions may also be included to capture the effects of natural disturbances such as wildfire, insect outbreaks, or wind events, although natural disturbances are generally modeled as predetermined proportions of forest impacted rather than management choices. A "no management" prescription option for each stratum is also included to describe the trajectory of vegetation change in the absence of both management and disturbances.

Constraints in a forest planning model typically reflect the limitations on implementing management activities. They are often built to comply with the standards, guidelines, and suitability of lands described in the forest plan (USDA Forest Service, 2008). For example, there may be limits on the amount of prescribed burning that can be implemented in a year, or a desirable amount of wildlife habitat to maintain through time. Another common constraint is the sustainability of timber harvest levels through time (called non-declining yield). Under the 2012 rule, forests are expected to act within its fiscal capability, i.e., stay within budget. Therefore, the total cost across all management activities in a given time period can be included as a constraint that should not be exceeded.

The model's objective function is a measure of solution quality, and often represents the primary objective of forest management. In USFS planning under the older 1982 rule, often the objective was the maximum total present net value from all planned forest management activities through a planning horizon. Under the 2012 rule, the objective function value is typically the minimum departure from the vegetation desired conditions in a forest, which represents the ecological objectives of the forest plan. Objective functions can also be used for exploratory information, such as to identify the most amount of timber that can be sustainably harvested in the near future.

Once the basic information about a forest is defined, managers must then coordinate the management of all forest strata through multiple periods of the planning horizon to achieve the primary objective while meeting all the management constraints. This complex problem is why DSS models are often needed to inform the selection process.

3. PRISM architecture

3.1. Overview

PRISM is an optimization DSS built on Linear Programming (LP) models. LP model formulations have been widely used for the design and analysis of forest management strategies. Two common LP formulations used in forestry are named "Model I" and "Model II" (Gunn, 2007; Davis et al., 2001; Martin et al., 2017). Both formulations are designed to allocate management activities across a forest landscape over multiple periods of a planning horizon. A detailed review of both formulations is beyond the scope of this study but can be found in many past publications in forestry literature (Martin et al., 2017; McDill et al., 2016; Belavenutti et al., 2018). Model I and Model II formulations have been applied in forest planning DSSs such as FORPLAN, SPECTRUM or WOODSTOCK (www.remsoft.com). PRISM allows users to implement both Model I and Model II formulations. Descriptions and illustrations of the Model I, Model II, and the PRISM hybrid model will be described in next sections.

We developed PRISM as an open-source software and made it accessible under the terms of GNU-GPL3, a free copyleft open-source software license. The GNU-GPL3 license allows users to freely download, use, distribute, and modify PRISM; the "copyleft" clause requires that derivative works from PRISM (i.e., its modified program and source-code) must be released to the public under the same GNU-GPL3 license. The open-source nature of PRISM makes the software inexpensive to use, update, and maintain.

The PRISM software includes three subsystems: a *database subsystem* for managing forest-specific data; a *modeling subsystem* for designing forest planning problems as LP models and solving those models; and an *output subsystem* for examining model solutions (Fig. 1). The three subsystems are wrapped into a GUI to facilitate interaction between users and the software. Many open-source add-ins (e.g., icons, tools, and libraries) are used to build the GUI and provide functionality for the subsystems. Detailed design and functionality of the PRISM's subsystems are described in the next sections.

3.2. The database subsystem

The database subsystem facilitates storing, manipulating, and examining forest planning input data in a relational database. Databases are displayed hierarchically with a "tree" structure in the GUI of this subsystem (Fig. 2A). A forest database is comprised of two key information: strata inventory and yield tables (i.e., prescriptions). Forest attributes are used to delineate forest strata and project forest condition changes over time for each stratum under the impact of different prescriptions. There is no restriction on the number of attributes that can be included in a forest database. PRISM includes an import function that loads information from delimited text files (i.e., csv files) and store them as database's tables. Delimited file specifications are defined in the PRISM user manual (link available in the **Software Availability** section). A noteworthy data management feature is that users may include multiple sets of strata and yield tables in the database to distinguish different forest plan alternatives.

SQLITE serves as the engine in PRISM to support many database operations including SQL (Structured Query Language) commands. A user-friendly feature of PRISM is it includes prewritten SQL commands (i.e., queries) designed to manipulate and examine forest data (Fig. 2B). So, most users do not need to learn SQL to do basic data exploration and manipulation. For additional flexibility, users may write their own SQL queries through the database subsystem GUI. User-defined queries may be saved to the PRISM SQL library that will be added to the dropdown query menu (Fig. 2B) for future usage.

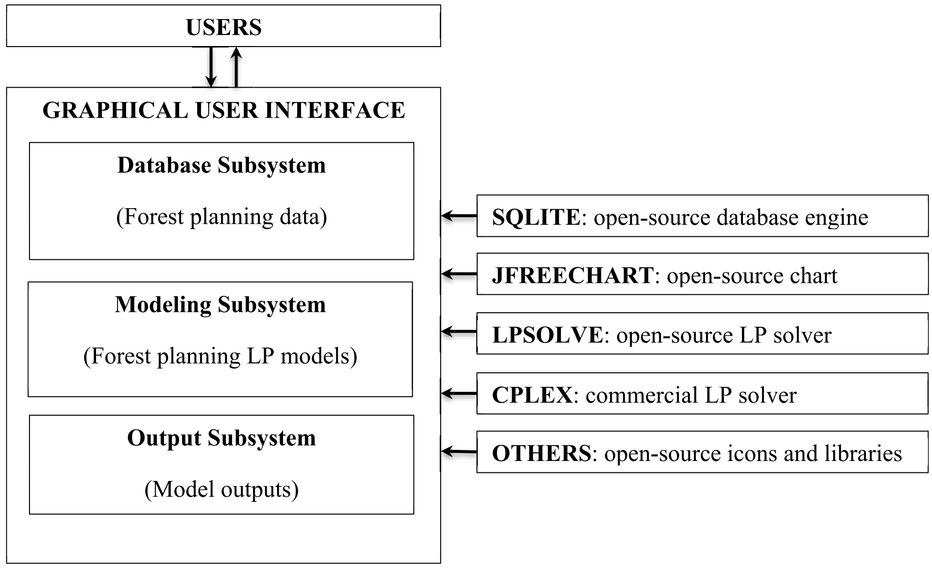

Fig. 1. Architecture of PRISM.

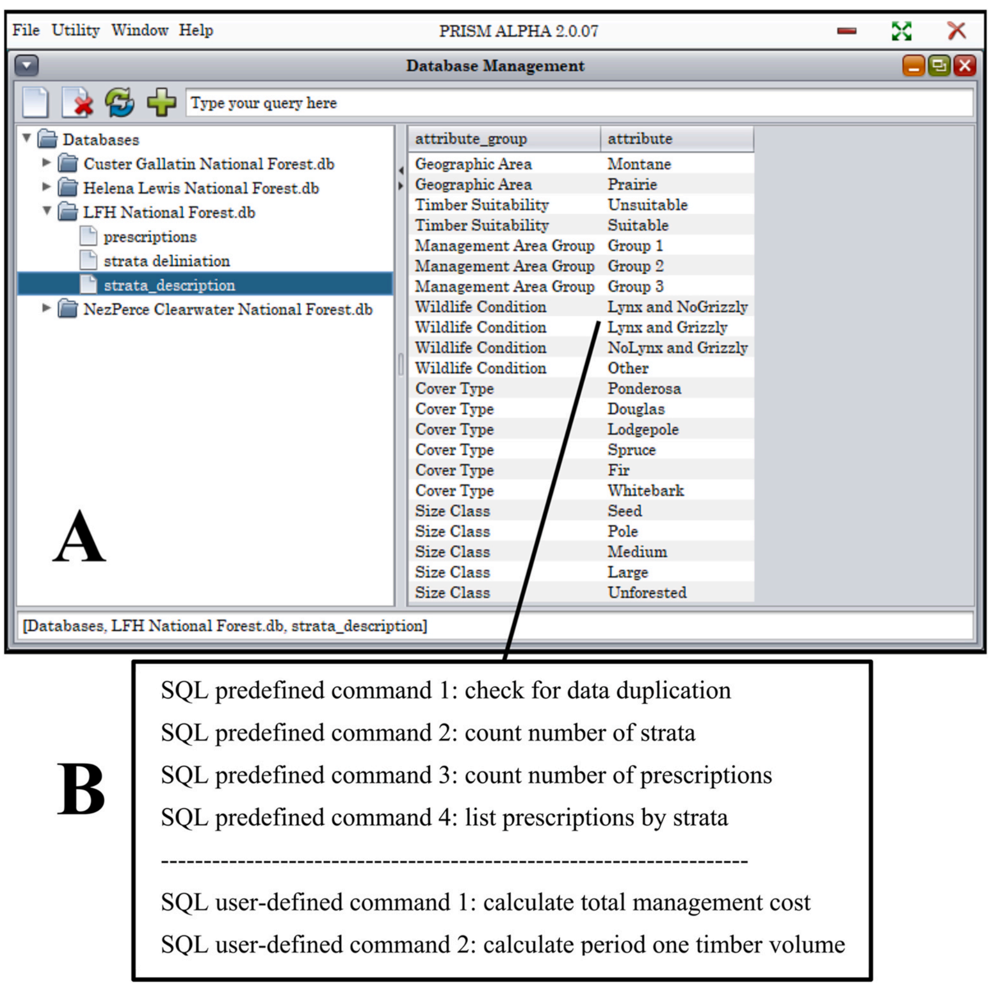

Fig. 2. PRISM's database subsystem. **A:** Database subsystem window that lists available databases and tables; **B:** Examples of predefined and user-defined SQL commands in the database subsystem accessed through right-click menu options.

3.3. The modeling subsystem

The modeling subsystem reads and validates forest strata and yield tables data from the database subsystem, allows users to design forest planning problems, and calls an LP solver to determine the optimal management strategy (Fig. 3). A modeling subsystem GUI is provided to

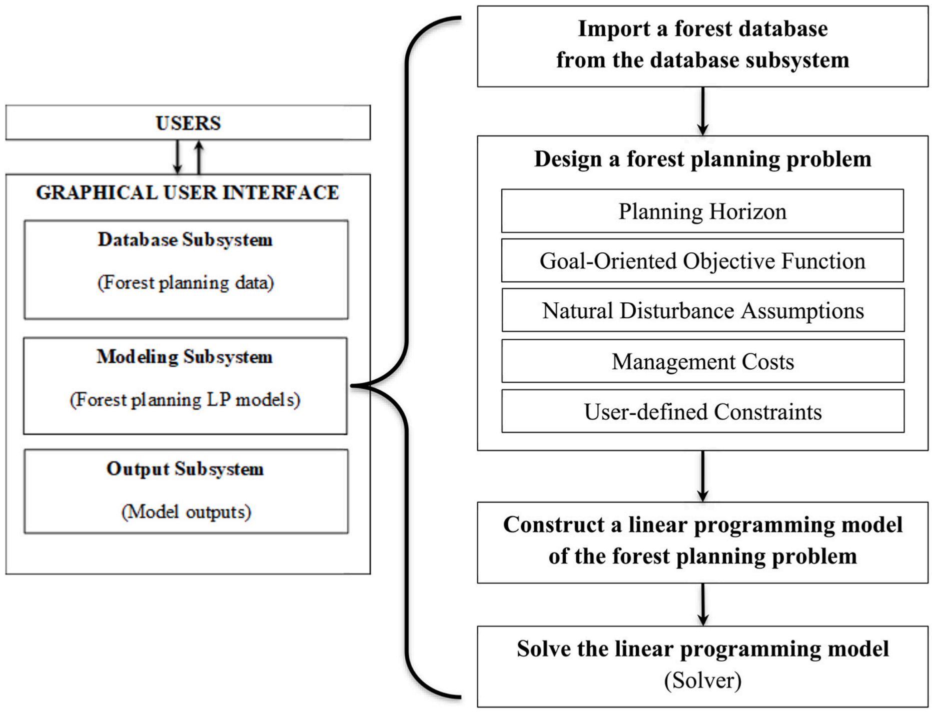

Fig. 3. Diagram of the PRISM's modeling subsystem.

help set up the following model components:

- *Planning horizon*: How many time periods are in the projection and how long is each period?
- *Goal-oriented objective function*: What is the main objective of the forest, and how do you measure success (e.g., movement towards desired conditions)?
- *Model formulation*: PRISM automatically decides whether to use a Model I, a Model II, or a combination thereof. The decision of which model to use is based on the design of management prescriptions in the input database.
- *Natural disturbances*: How many types of natural disturbances will be defined in the model, how likely are they to occur, and what are their impacts?
- *Management costs*: What are the financial costs associated with different management activities?
- *User-defined constraints*: What are the management requirements? This encompasses a wide range of options including timber production and sustainability targets, restrictions on management activities, wildlife habitat area requirements, budget limitations, desired forest species distribution, etc.
- *Solver*: The LP solver used to identify the optimal management strategy.

Sections from 3.3.1 to 3.3.6 will highlight some of these features in the PRISM's modeling subsystem. More detailed information can be found in the user manual of PRISM available for download from the website provided in the **Software availability** section.

3.3.1. Goal-oriented objective function

In PRISM, management problems are defined using a non-preemptive goal programming approach. This approach was also used in other software such as a project-scale forest planning tool called NED (Twery et al., 2000) and its successor NED-2 (Twery et al., 2005). In this approach, forest management goals are optimized simultaneously and weighted according to their importance. For example, desired future forest species composition can be constructed as a series of goals for each unique vegetation condition. Each goal specifies a target level as either an absolute value or a numeric range between the minimum and maximum acceptable levels. The importance of each goal is defined with a user-defined weight, which is used to calculate a penalty if the goal is not met. When a goal cannot be achieved (under-achievement or over-achievement compared to the target level), the weight is multiplied by the deviation from its target level to determine the penalty. The objective function of a PRISM model is to minimize the sum of penalties across all goals. PRISM can also examine single-objective problems such as finding maximum timber volume. In this case, users simply set an arbitrarily large timber volume as the only goal in the formulation with a non-zero weight. Assuming the goal is larger than the volume determined by the solution, the solution will represent the maximum possible volume. The detailed goal programming formulation implemented in PRISM is described in the **Appendix**. Allowing users to easily model unlimited number of goals is a major contribution of PRISM design that makes it suitable for forest planning with multiple objectives toward desired future conditions.

3.3.2. Model formulation

Model I (Fig. 4A) and Model II (Fig. 4B) are the two well-established LP model formulations used in forest management and planning. A major difference between these two models is the consistency through time of forest management strata responding to regeneration events such as timber harvest or lethal disturbance. Regeneration events result in "bare land" that has associated management choices such as natural regeneration or planting to a particular tree species. A Model I formulation allocates the area in each stratum to management prescriptions through the entire model duration (or planning horizon). In contrast, a Model II formulation tracks the area in each stratum assigned to prescriptions that span the length of time between forest regeneration events (in contrast to the entire planning horizon). Each forest stratum used in Model I retains a constant area through the planning horizon. The area of Model II strata change through time as portions may become bare lands at every period due to harvest or disturbance events; bare land areas originated in the same period can be combined to create new strata for regenerating different vegetation species.

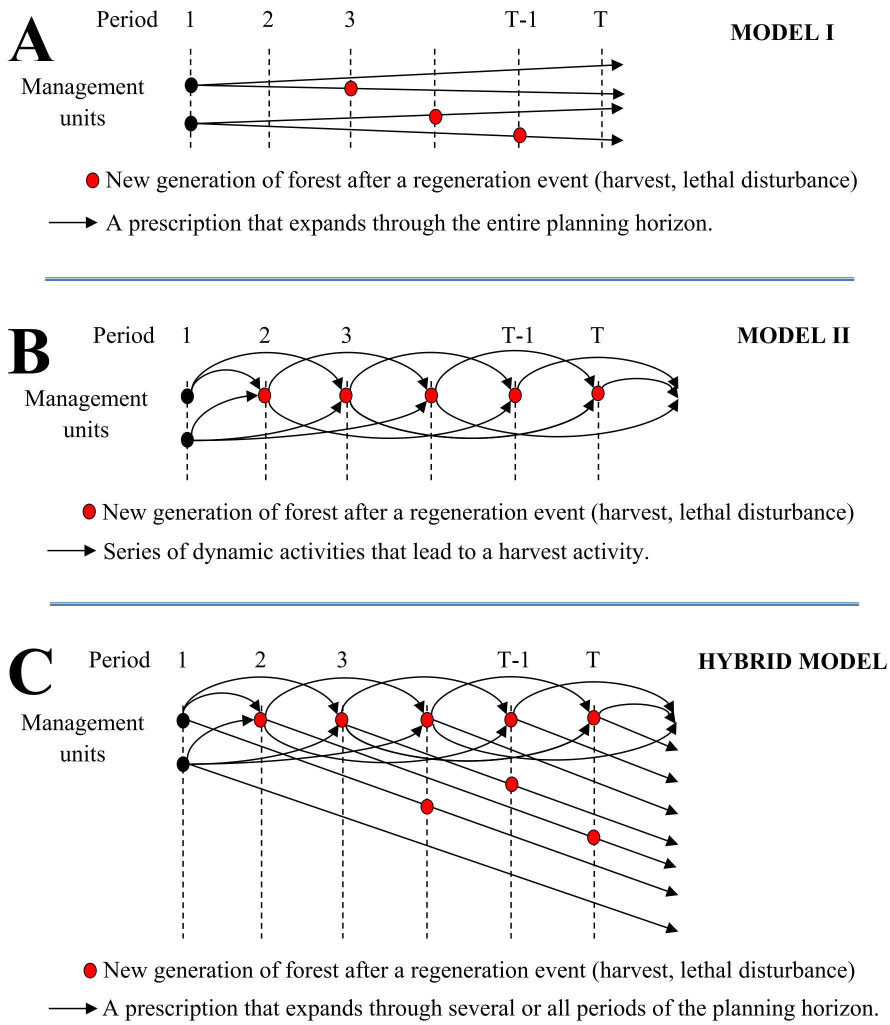

Fig. 4. Illustration of some forest management models. **A:** Model I. **B:** Model II. **C:** Hybrid Model. Note that while Model I prescriptions have regeneration events, there is only one way in and one way out; thus, the integrity of the stratum is preserved through time. Model II prescriptions aggregate bare lands created after regeneration events and form a new stratum to evaluate from that point forward.

PRISM uses a hybrid model structure to define management options for forest strata (Fig. 4C). This has several advantages: 1) Users do not have to design all prescriptions to expand through the entire planning horizon (typical Model I design). For example, the same Model II harvest prescription may be created and used repeatedly to manage a stratum through any 5 consecutive periods (1–5, 2–6, 3–7, etc.) of a planning horizon. 2) PRISM model allows flexible strata management because users can define rules for merging strata that regenerate in the same time period. For example, users can require that only the mixed-conifer and lodgepole pine strata harvested in the same period may be merged. They can also choose specific vegetation (available from those Model II prescriptions) for regeneration, such as the merged mixed-conifer and lodgepole strata after harvest may be naturally regenerated as mixed-conifer, or can be planted (for a cost) to lodgepole. 3) Model selection requires no user effort because PRISM automatically implements the appropriate model (I or II or both) for each stratum based on its available prescriptions.

3.3.3. *Ability to use forest attributes in the problem formulation*

PRISM has a user-friendly GUI to account for both static and dynamic forest attributes (Fig. 5) in the LP formulation. Specifically, the GUI allows users to select forest strata based on static attributes (Fig. 5A), specific periods of interest (Fig. 5B), and dynamic attributes (Fig. 5C). For example, the checked boxes in Fig. 5 would select strata in the "Montane" area with "Lynx habitat" where the initial vegetation is "Ponderosa" with "Medium" and "Large" tree sizes, given that in the second period the trees in those strata are at "age class 3" and the density is within "300–600 trees per acre". The selected strata can then be used to allocate natural disturbances, assign management costs, and define constraints representing various management restrictions on area, or cost, or forest attributes (e.g., carbon storage, wildlife habitat value) associated with those strata across time as described in the next sections of this paper.

3.3.4. *Natural disturbances*

PRISM can model both non-lethal disturbances (NDs) and lethal disturbances (LDs). ND events are assumed to damage a portion of the trees in a stratum, while LD events are lethal to most or all trees in the affected stratum area. After an LD event, a newly regenerated forest starts a new life cycle.

Non-lethal disturbances are modeled at the prescription generation stage with Model I prescriptions built to track the effects of predefined sequences of ND events through the entire planning horizon (e.g., mountain pine beetle outbreaks every 30 years). PRISM's GUI allows users to assign appropriate prescriptions with ND events to a portion of each stratum. In contrast to NDs, lethal LD disturbances are modeled

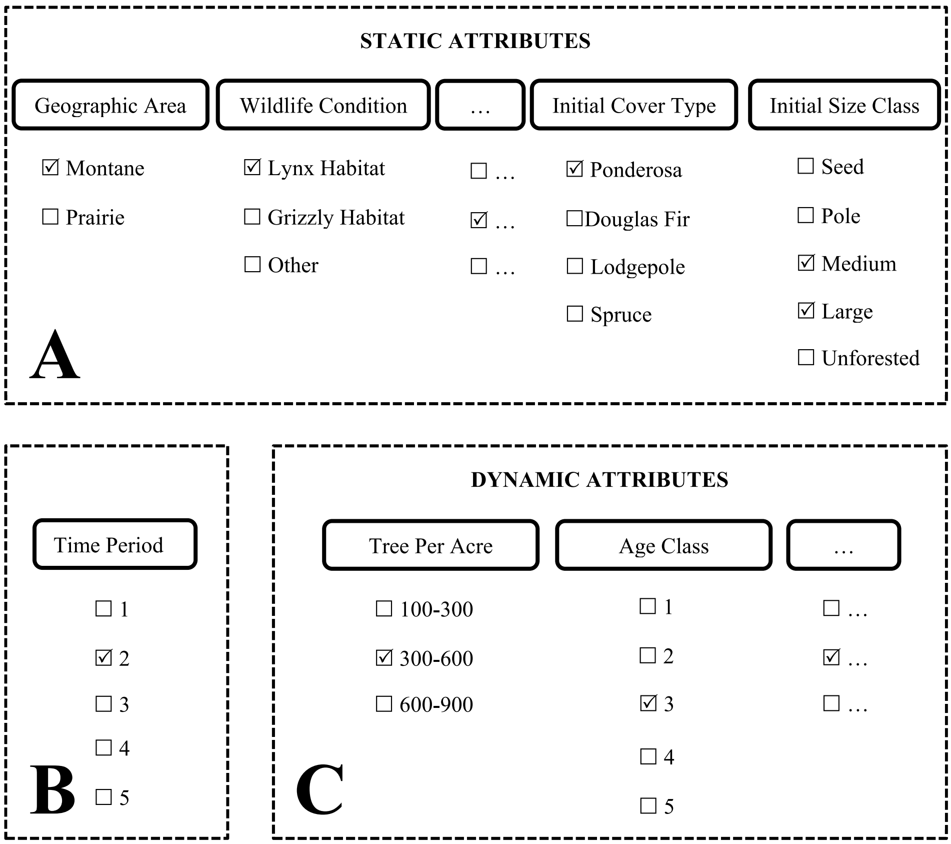

Fig. 5. GUI components allowing users to interact with all forest strata attributes in the database. The triple dots (…) represent the ability of PRISM to include many other attributes into the GUI. **A:** *Static attributes.* **B:** *Planning periods.* **C:** *Dynamic attributes* with user-defined discrete values (age classes) and range of values (trees per acre).

without using prescriptions. LD events are defined directly through the PRISM GUI according to what strata the events occur in, occurrence times, proportion of each stratum impacted, corresponding regenerated vegetation, and regeneration trajectories after each LD event (i.e., appropriate Model II prescriptions to manage the new bare land strata after LDs).

PRISM models the impacts from NDs and LDs in different ways. NDs are modeled deterministically as fixed percentages of the total stratum area disturbed; while LDs are modeled as stochastic percentages. The deterministic approach simply uses the user-defined average probability in each stratum during each planning period. The stochastic approach draws random disturbance levels across the planning horizon based on the user-defined probability distributions.

3.3.5. Management costs

PRISM also has a GUI to help users define management activity costs including the per-area treatment cost, cost associated with a dynamic attribute (such as timber volume), and regeneration cost that varies by specific vegetation types to plant. Costs can also be built as fixed values into prescriptions in the forest database. However, defining costs directly through the PRISM's GUI makes it easier to change and test different assumptions without the need to modify the database. For instance, two strata managed by the same prescription may be defined in the GUI to have different treatment costs if road building is required to access one stratum but not the other.

3.3.6. User-defined constraints

PRISM is designed to efficiently develop and test constraints in a forest planning model. Constraints are categorized as "non-relational" and "relational", each with a separate interface design. Some relational constraints model the flows of timber, cash or other management elements across time and are called "flow constraints". Non-relational constraints define direct limitations on area, cost, or other quantified forest conditions or production levels (Fig. 6). Examples include a minimum amount of wildlife habitat area, a maximum amount of budget spending, or a limit on the timber volume that can be generated from clearcuts etc. Relational constraints can be used to model other relationship requirements (Fig. 7). For example, they can be used to limit timber volume fluctuation between two consecutive time periods, control the ratio of even-aged to uneven-aged forest treatments within the same period, or define ending-period inventory of standing volume relative to the average standing inventory of the previous several periods. The combination of the two constraint categories creates a flexible modeling framework to describe a forest management problem with various management requirements. More comprehensive definitions and explanation of these constraints can be found in the "PRISM Goal Programming Formulation" section of Appendix.

3.3.7. Solution options

PRISM supports running multiple models in two available batch modes. The sequential mode solves models one after another, while the parallel mode implements a Java concurrency technique (multi threads) to enable solving many models simultaneously. Presently, the PRISM interface provides two solver options: LPSOLVE (free open-source solver) or CPLEX (commercial solver requiring a license). LPSOLVE can handle small to medium size LP problems. Large models, such as those with more than one million variables, may require the more powerful CPLEX solver. Those two options make PRISM useful for both academic training (lower cost) and industrial applications.

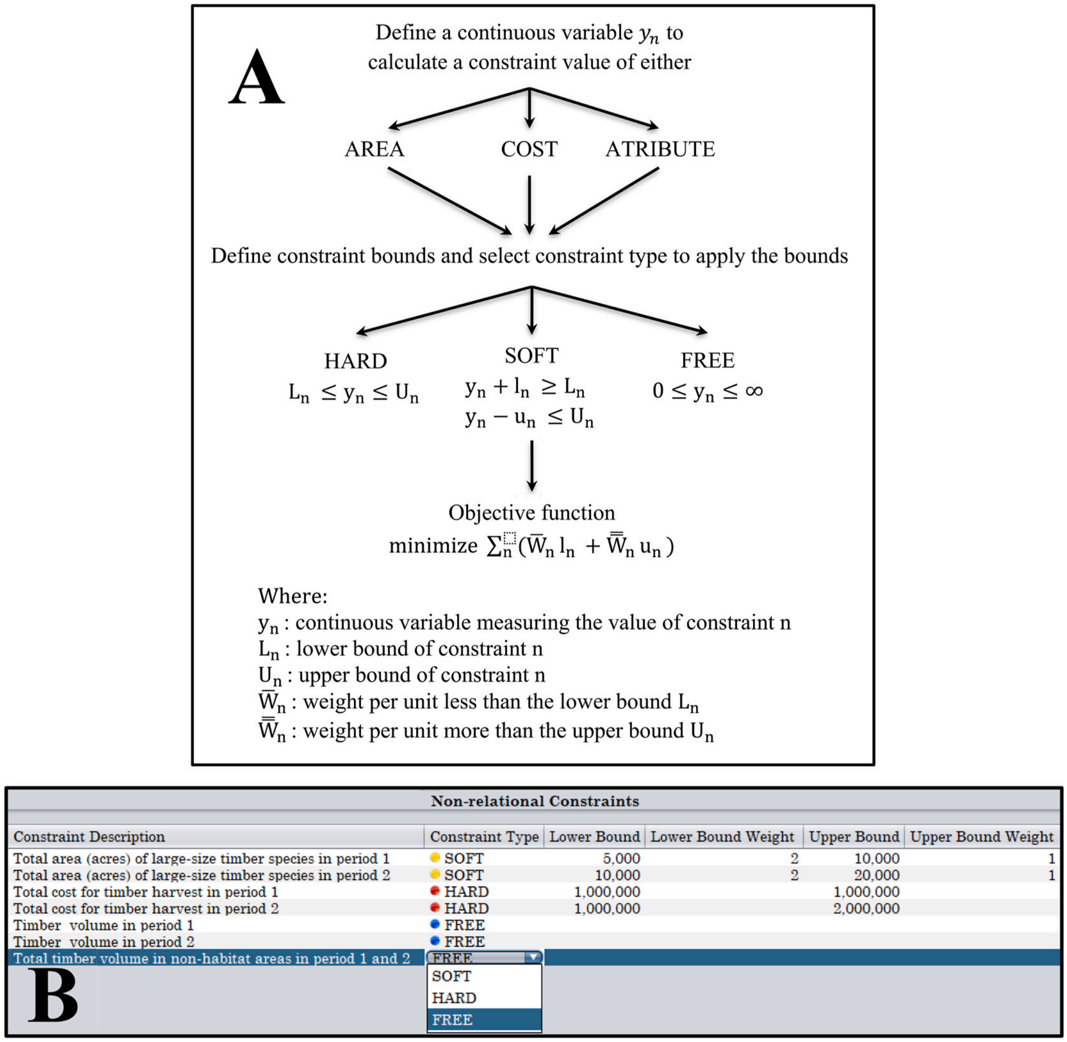

Fig. 6. Non-relational constraints in PRISM. Each non-relational constraint is created by using a single y_n variable and its bounds. Constraint type (HARD, SOFT, FREE) represents three different options to apply the bounds to the variable. More explanation of non-relational constraints is presented in Appendix. **A:** Process of building non-relational constraints. **B:** A screenshot of the PRISM GUI for building non-relational constraints.

3.4. The output subsystem

After an LP model is solved, the output subsystem gives users a variety of options to manage and visualize the model solution through both standard and custom output options. Standard outputs include the following:

- Solution metrics such as solution time, objective function value, number of decision variables, and number of constraints.
- Information about decision variables in the optimal solution including their values and reduced costs.
- Information about constraint levels at the optimal solution including dual and slack values.
- A query framework that includes predefined SQL commands to summarize model outputs.
- Reports and graphical displays of the model outputs such as selected management activities for each stratum and constraint levels. The output subsystem uses an open-source chart library (JFREECHART) to visualize the output data by building multiple types of charts (e.g., pie or bar charts in Fig. 8). The data visualization feature can help users get quick insights from the model results.

In addition to the standard outputs, PRISM allows for the development and storage of customized SQL commands, as well as GUI-enabled query tools to examine the optimal solution without SQL. Outputs from SQL queries and the non-SQL GUI-Tool queries can be exported to external software for supplemental analysis and display options.

4. PRISM application and performance

Since 2016, PRISM has been used to support land management planning for several US national forests. These include the Custer-Gallatin National Forest (CGNF) in southern Montana and western South Dakota, the Helena-Lewis and Clark National Forest (HLC) in north central Montana, and the Nez Perce-Clearwater National Forest (NPC) in north central Idaho. The land management plan for each forest required a detailed analysis of multiple management alternatives to comply with the direction in the 2012 Planning Rule and the National Environmental Policy Act. For each national forest, six management alternatives were developed based on the forest's internal assessment and public comments, and represent a range of different management strategies and outcomes. Alternatives may differ by management emphasis (Appendix Table A1) associated with different levels of contribution to social and economic sustainability (Appendix Table A2). Differences among alternatives are due to local and national stakeholder group preferences that prioritize certain social and economic benefits over the others. For example, the CGNF designed alternative D for the greatest contributions to clean water, fish and wildlife conservation,

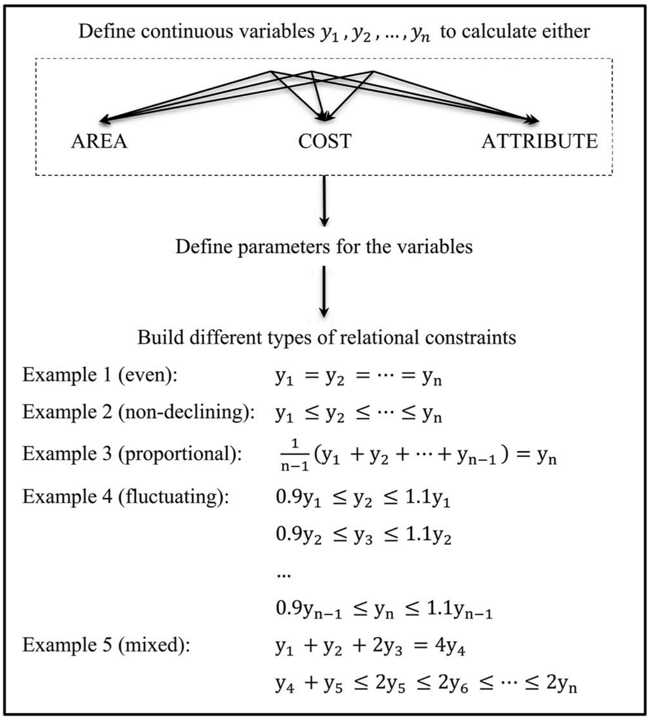

Fig. 7. Process of building relational constraints in PRISM. Relational constraints are formulated as a relationship between two or more y_n variables. More explanation of relational constraints is presented in the Appendix.

wildfire and fuel management, cultural-historic-tribal resource protections, and scenery while alternative E was designed for the greatest contributions to clean air, timber production, income, and job opportunities. Alternatives B, C, and F were aimed at balancing a mix of contributions across all stakeholder groups (Appendix Table A2).

PRISM was used to analyze the long-term outcomes and effects of the alternatives for each of these plans. Each plan alternative is built as a distinct PRISM model. The objective for each model was to schedule vegetation treatments to move the forest toward desired conditions within a 15-decade planning horizon. Through weighting the benefits and trade-offs of each alternative over the life of the forest plan, forest managers can make informed decisions on which alternative to implement in the plan. Comprehensive PRISM results for each plan are available on the USFS plan revision web pages (USDA Forest Service, 2020a; USDA Forest Service, 2021; USDA Forest Service, 2019). Sections below describe the application and performance of the software.

4.1. PRISM application

Table 1 summarizes the PRISM model inputs from the three different national forests. For each forest, strata were delineated based on various static spatial attributes such as geophysical, political, and management area designations. Vegetation attributes were derived from the Northern Region Existing Vegetation Mapping Program (USDA Forest Service, 2020b). Prescriptions and associated forest dynamic attributes were derived from projecting Forest Inventory and Analysis data (Wurtzebach et al., 2020) of each forest into the future with the Forest Vegetation Simulator software (Crookston and Dixon, 2005).

The three forests used a broad range of forest management cost categories including timber sale preparation and administration, reforestation, timber stand improvement, road reconstruction and administration, prescribed burning, weeds treatment, and other surcharges. Specific cost values varied by management activities, static attributes such as physical location, and dynamic attributes such as vegetation type and density.

Three types of natural disturbances were modeled on the three forests. Two disturbances were non-lethal (mixed severity wildfire and severe insect infestation) and the other was lethal wildfire. The USFS relied on historical burn severity and insect outbreak data to identify the expected amounts of future natural disturbance events. For example, the CGNF assumed a 50-year interval for mixed severity wildfire, 60-year interval for severe insect outbreaks, and between 9000 and 11,500 acres per year of lethal fire. Disturbance amounts were put into PRISM as fixed percentages of area that would be disturbed in every decade.

A wide variety of constraints were built to ensure compliance with laws, policy, and the management direction of the proposed plan alternatives. Constraints were classified into several groups including: *budget constraints* (budget limits on management activities), *harvest policy constraints* (non-declining volume and other restrictions on timber harvest), *dispersion of openings constraints* (distribution of management activities across strata to prevent habitat fragmentation at any point in time), *prescribed burning constraints* (the upper limit of prescribed burning area to meet air quality standards, operational capacity, and weather limitations), *wildlife habitat constraints* (to help ensure sustainable habitat levels), and *treatment constraints* (to meet fuel reduction area targets and other management requirements). Those constraints were designed to meet management requirement targets (both under and over achievements of the required management targets are not allowed). Another group of constraints was built to model the desired area distribution (including both the minimum and maximum levels) for different forest vegetation types and structure/size classes. These constraints allow for deviations from the desired vegetation condition levels that were used as the basis for the goal-based objective function.

PRISM was used to generate the projected acres of management activities and annual timber volume (see Appendix Table A3 for the CGNF reports) included directly in each forest's land management plan. PRISM was also used for sensitivity and trade-off analyses to evaluate the effect of various plan components (expressed as constraints) on the forest's ability to meet desired conditions. For example, by systematically eliminating constraints from the model and measuring the maximum volume achieved in the first decade, the CGNF found that the non-declining timber volume constraint had the most impact on management results (Appendix Table A4). Another analysis was conducted by the HLC where PRISM was used to quantify trade-offs between different budget allocations and identify whether the budget was not a limiting factor.

4.2. PRISM performance

To evaluate PRISM performance, models from the three national forests were solved by using IBM's ILOG-CPLEX 12.8 on a 64-bit workstation equipped with a dual-core 2.5 GHZ processor and 32 GB of RAM. For each national forest, six PRISM models associated with the six plan alternatives developed for that forest were solved. PRISM was able to solve every existing model and reported corresponding results within 30 min. The average results are presented for each forest in Fig. 9. Model size and complexity vary for each forest, reflected by number of variables and constraints in the model. Model performance, reflected by total run time and memory usage, is different depending on model size and complexity.

Models developed for the CGNF, HLC, and NPC respectively have increasing sizes as measured by the number of variables and the number of constraints (Fig. 9A). Larger models require more computer memory to store and process the model data and solve the associated LP problems (Fig. 9D). Larger models typically require longer total run times (Fig. 9C). Here, the total run time is the sum of: 1) *input processing time*, the time it takes PRISM to read the user-inputs from the GUI and formulate the LP problem, (2) *solution time*, how long it takes CPLEX (the default solver) to solve the LP model, and (3) *output writing time*, how long it takes to prepare and import model results to the output subsystem. There is a positive relationship between model size (Fig. 9A) and

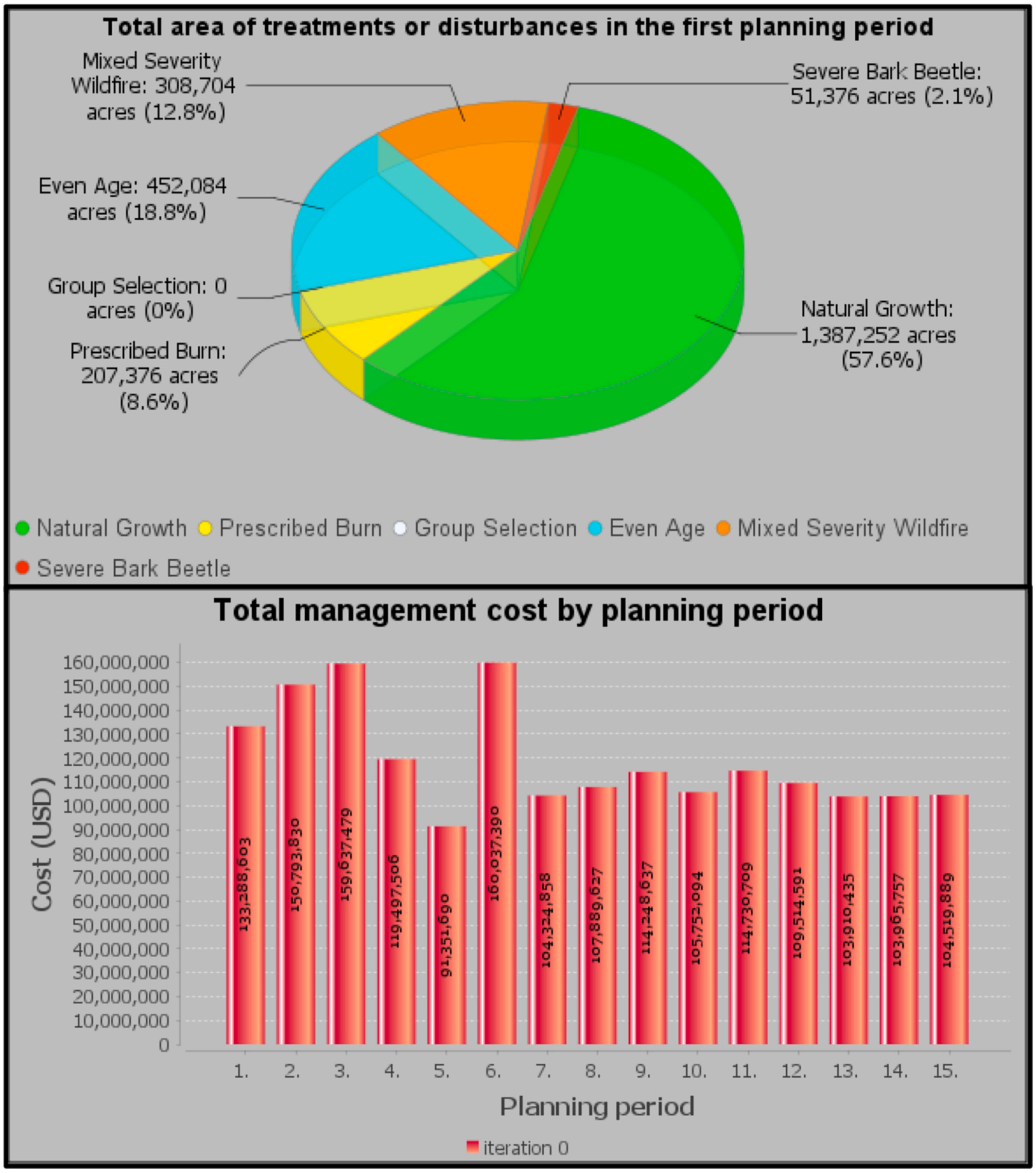

Fig. 8. Screenshots of pie and bar charts captured directly from the PRISM software.

Table 1
Forest planning inputs of the three national forests summarized across alternatives.

Forest planning input	CGNF	HLC	NPC
Number of management alternatives in the forest plan	6	6	6
Planning horizon (number of periods)	15	15	15
Forest area (thousand acres)	1952	2407	3781
Number of strata (varies by alternative)	3397–4174	3481–4045	3749–3749
Number of prescriptions	3820	5268	2836
Number of static forest attributes used to build strata	46	44	62
Number of dynamic forest attributes included in prescriptions	69	121	55
Number of non-lethal disturbance types	2	2	1
Number of lethal disturbance types	1	1	1

the input processing time and the output writing time (Fig. 9C). The number of non-relational constraints is often much larger than the number of flow constraints for each forest (Fig. 9B). Both non-relational constraints with tighter bounds and flow constraints can make an LP model difficult to solve. For example, even though the NPC models are larger (Fig. 9A), they take less time to solve (Fig. 9C), likely because they have fewer flow constraints (Fig. 9B). Sensitivity tests could help define the impacts of different constraints on solution time, but this is not our focus in this study.

5. Discussion and conclusions

PRISM is a new forest planning DSS developed through an agreement between the USFS Northern Region and Colorado State University. It has been used by several US national forests to support the development of forest plans, but its functionality is flexible enough to support planning efforts by other forestland owners and managers as well.

PRISM has several strengths that may make it a preferable tool for forest management analysis. First, it is available through a no-cost open-

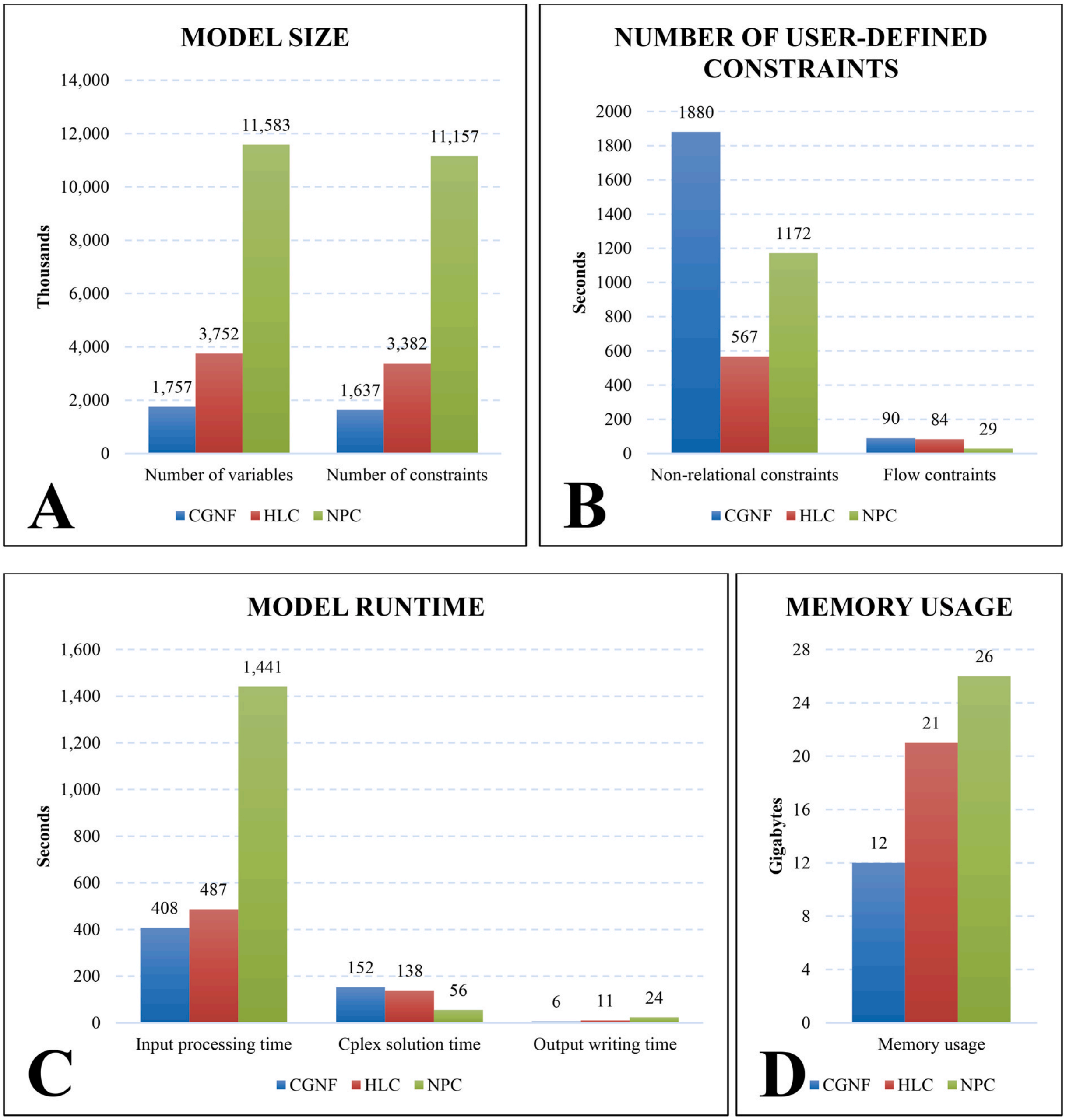

Fig. 9. PRISM performance based on models from three different national forests. **A:** Model size. **B:** Number of user-defined constraints. **C:** Model runtime. **D:** Memory usage.

source license. Its database management is supported by the open-source SQLITE database engine, and visualization options are provided by the open-source JFREECHART library. While a powerful commercial solver may incur a licensing fee, smaller problems may be solved by the open-source solver LPSOLVE integrated in PRISM. Another strength of PRISM is its architecture to support rapid model development and testing process. PRISM allows users to delineate a forest into many strata, incorporate multiple choices for strata management, and define various model components (natural disturbances, management costs, management constraints) directly through the GUI. GUI features, such as the constraints section in particular that allows users to quickly switch constraint types, facilitate solution space exploration and pinpointing infeasibilities. The ability to run models in batch mode to quickly compare the results is another advantage of PRISM. It saves significant time for analysts to run multiple scenarios and present results to decision makers. Finally, the source code for PRISM is stored and managed through a GIT control open-source website that is convenient for future software maintenance and expansion.

Natural disturbance and associated uncertainty create challenges for determining an optimal management schedule based solely on using optimization DSS. To overcome these challenges, the three US national forests discussed here integrated PRISM with other DSS software to form an analytical system. Specifically, at the front end, the Forest Vegetation Simulator (Crookston and Dixon, 2005) was used to determine vegetation growth trajectories for forest strata including the effects of non-lethal disturbances. This information made the yield tables that served as inputs for PRISM. PRISM was then used to schedule forest management activities through time under deterministic impacts from both non-lethal and lethal disturbances (lethal disturbances are defined through PRISM's GUI). The management schedule from PRISM was in turn analyzed with multiple stochastic disturbance scenarios by a state-transition model, SIMPPLLE (Chew et al., 2004) to evaluate the management effectiveness in the context of future uncertainty. In such an iterative analytical system, parameters of each model can be adjusted based on the outcomes of another to gain a better understanding of the interaction between management and ecological processes (Chick et al., 2019).

Integrating multiple DSS models is a time-consuming process, as it requires that each model be parameterized, calibrated, and run. PRISM holds the potential for improved future efficiencies if it is modified to simulate stochastic disturbances with better details. This would allow simultaneous evaluation of management and ecological processes in a single PRISM model. Multiple PRISM model runs could produce a range of possible management activities and ecological trajectories. Some further options worth investigating include stochastic programming models (Wei et al., 2014) that can provide a solution by simultaneously accommodating many possible future scenarios, a rolling horizon mechanism (Savage et al., 2010) that evaluates forest management strategy by an adaptive management approach, or simply running multiple scenarios of the same model with stochastic disturbances. If developed, these capabilities would be another example of how PRISM has improved upon the previous generation of USFS planning models.

PRISM has the potential to be maintained and expanded through time. It was developed using an open-source development platform. This form of collaboration allows more flexibility for both the end users and the developers to continuously maintain and improve PRISM. It also encourages participation from other potential users or developers to the software's future development. The open-source nature of PRISM enhances its chance to be adopted by managers and experts in the domain of forest management and planning.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Link to source code is included in the manuscript.

Acknowledgements

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Appendix

Prism Goal Programming Formulation

PRISM formulates the planning problem as a Linear Programming (LP) model. Formulating a LP model requires identifying a set of decision variables for each stratum to represent where, when, and what management prescriptions may be chosen. In PRISM, decision variables are associated with forest strata through information stored in yield tables. Decision variables can be incorporated into constraints to capture various assumptions, relationships and requirements of managing the forest. They may also be incorporated into the model's objective function that minimizes the total deviation from multiple forest management goals. The solution determined by the PRISM model indicates a forest management plan that leads to the best objective function value. The mathematical expression of the PRISM LP formulation is provided here:

Objective

$$min: \sum_{n} \left(\overline{W}_n \, l_n + \overline{\overline{W}}_n \, u_n \right) \tag{1}$$

Constraints set 1: non-relational constraints

$$y_n + l_n \geq L_n \quad \forall n \in C^{SOFT} \tag{2a}$$

$$y_n - u_n \leq U_n \quad \forall n \in C^{SOFT} \tag{2b}$$

$$y_n \geq L_n \quad \forall n \in C^{HARD} \tag{3a}$$

$$y_n \leq U_n \quad \forall n \in C^{HARD} \tag{3b}$$

$$y_n \geq 0 \quad \forall n \in C^{FREE} \tag{4a}$$

$$y_n \leq +\infty \quad \forall n \in C^{FREE} \tag{4b}$$

Constraints set 2: relational constraints (flow constraints)

$$\sum_{n \in C_{m,\ k+1}} y_n \geq \vec{L}_m \sum_{n \in C_{m,k}} y_n \quad \forall m, k \in [1, K_m - 1] \tag{5a}$$

$$\sum_{n \in C_{m,\ k+1}} y_n \leq \vec{U}_m \sum_{n \in C_{m,k}} y_n \quad \forall m, k \in [1, K_m - 1] \tag{5b}$$

Constraints set 3: transitional constraints

$$\sum_{i\in I_x} x_{s,i,1} + \sum_{s'}\sum_{i\in I_x} \overline{x}_{s,s',i,1} = V_s \quad \forall s \tag{6}$$

$$(1-P_x)x_{s,i,t} = x_{s,i,t+1} \quad \forall s, i \in I_x, t \in [1, T-1] \tag{7}$$

$$(1-P_{x'})x'_{s,i,t,a} = x'_{s,i,t+1,a+1} \quad \forall s, i \in I_{x'}, t \in [1, T-1], a \tag{8}$$

$$\sum_{i\in I_x} P_x P_{s,s',x} x_{s,i,t} + \sum_a \sum_{i\in I_{x'}} P_{x'} P_{s,s',x'} x'_{s,i,t,a} = f_{s,s',t} \quad \forall s, s', t \in [1, T-1] \tag{9}$$

$$\sum_s f_{s,s',t} + \sum_s \sum_{i\in I_x} \overline{x}_{s,s',i,t} + \sum_s \sum_{i\in I_{x'}} \sum_a \overline{x}'_{s,s',i,t,a} = \sum_{i\in I_{x'}} x'_{s',i,t+1,1} + \sum_{s''} \sum_{i\in I_{x'}} \overline{x}'_{s',s'',i,t+1,1} \quad \forall s', t \in [1, T-1] \tag{10}$$

Indexes

n	index of a non-relational constraint
m	index of a relational (i.e., flow) constraint
k	index of a flow's component
s, s', s''	index of a stratum based on user-defined static attributes such as physical location, timber suitability, etc.
i	index of a prescription. A prescription can capture the effects from various silviculture methods (i.e., natural growth, prescribed burning, group selection, even-aged harvest) or non-lethal natural disturbances (i.e., mixed severity fire, insect outbreak)
t	index of a planning period
a	index of an age-class

Parameters

L_n	lower bound of constraint n. ($L_n \geq 0$)
U_n	upper bound of constraint n. ($U_n \geq 0$)
$\overline{W}_n$	weight for one unit less than L_n. ($\overline{W}_n \geq 0$)
$\overline{\overline{W}}_n$	weight for one unit more than U_n. ($\overline{\overline{W}}_n \geq 0$)
K_m	number of components of flow m. ($K_m \geq 2$)
$\overrightarrow{L}_m$	lower bound of flow m. ($\overrightarrow{L}_m \geq 0$)
$\overrightarrow{U}_m$	upper bound of flow m. ($\overrightarrow{U}_m \geq \overrightarrow{L}_m$)
V_s	area associated with the existing stratum s at the beginning of planning horizon
P_x	(user-defined) loss rate due to lethal disturbances in the existing stratum area x
$P_{x'}$	(user-defined) loss rate due to lethal disturbances in the regenerated stratum area x'
$P_{s,s',x}$	conversion rate associated with P_x where $\sum_{s'} P_{s,s',x} = 1\ \forall s$
$P_{s,s',x'}$	conversion rate associated with $P_{x'}$ where $\sum_{s'} P_{s,s',x'} = 1\ \forall s$

Sets

C^{FREE}	set of FREE non-relational constraints
C^{HARD}	set of HARD non-relational constraints
C^{SOFT}	set of SOFT non-relational constraints
$C_{m,k}$	set of user-defined continuous variables that creates the component k of the flow m. For example, the flow m can be described by 3 components $k = \{1, 2, 3\}$ where each component may contain multiple continuous variables such as $C_{m,1} = \{y_1\}$; $C_{m,2} = \{y_2, y_3\}$; and $C_{m,3} = \{y_5, y_7, y_9\}$
I_x	set of prescriptions that are feasible for the existing strata variable x
$I_{x'}$	set of prescriptions that are feasible for the regenerated strata variable x'

Variables

$x_{s,i,t}$	area of the existing stratum s in the period t which would be managed by prescription i and the activity to be implemented in period t is **not** clear-cut
$\overline{x}_{s,s',i,t}$	area of the existing stratum s in the period t which would be managed by prescription i, the activity to be implemented in period t is clear-cut, and the regenerated landscape attributes identification would be s' in the next period $t+1$

$x'_{s,i,t,a}$ area of the regenerated stratum s in the period t at age class a which would be managed by prescription i and the activity to be implemented in period t is not clear-cut

$\overline{x}'_{s,s',i,t,a}$ area of the regenerated stratum s in the period t at age class a which would be managed by prescription i, the activity to be implemented in period t is clear-cut, and the regenerated landscape attributes identification would be s' in the next period $t+1$

$f_{s,s',t}$ total area of the stratum s subjected to lethal disturbances in period t and regenerated into the forest landscape identification s' at age-class one in period $t+1$

y_n a user-defined continuous variable to calculate a quantified level of either area (e.g., acres), or cost (e.g., dollars), or any numeric forest attribute (e.g., number of trees, harvested volume) associated with certain strata in certain time periods

l_n total units falling short of L_n ($l_n \in [0, L_n]$)

u_n total units exceeding U_n ($u_n \geq 0$): might not be true for the case when $\overline{\overline{W}}_n = 0$

Where

Decision variables use their indices to link to a specific row of a specific prescription in the input database (Fig. A1). PRISM uses modified Model I variables ($\overline{x}_{s,s',i,t}$ *and* $x'_{s,i,t,a}$) and modified Model II variables ($x'_{s,i,t,a}$ *and* $\overline{x}'_{s,s',i,t,a}$). Specifically, the time period (t) index is not necessary for Model I variables, but PRISM uses t to capture the area transition due to the effects of lethal disturbances in each planning period (note that lethal disturbance conditions, proportions, and effects are defined in the PRISM GUI instead of using prescriptions). While Model II variables often use the starting and ending period indices associated with each prescription, PRISM uses time period (t) and age class (a) as alternative indices with an equivalent effect. For example, a regeneration prescription to regrow vegetation in a bare land in 4 consecutive periods from 2 to 5 would be associated with the set of (t, a) = { (2,1), (3,2), (4,3), (5,4) }.

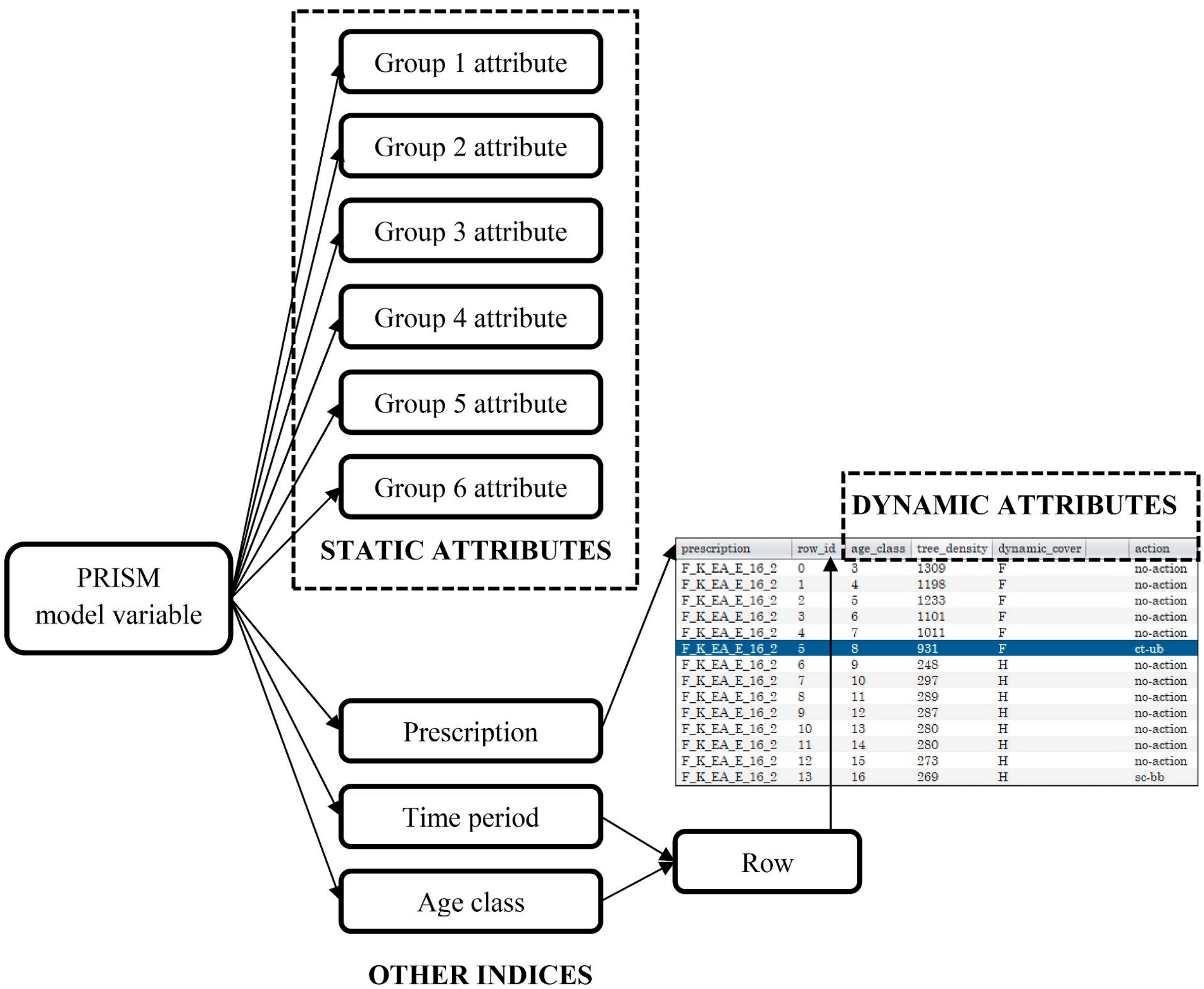

Fig. A1. Design of decision variables in PRISM. PRISM allow using up to six different user-defined groups of static attributes (integrated into the s index) to index a decision variable ($x_{s,i,t}$, $\overline{x}_{s,s',i,t}$, $x'_{s,i,t,a}$, $\overline{x}'_{s,s',i,t,a}$). Other indices ($i, t, a$) are used to link every variable to an exact row of an exact prescription in the database. That single row contains detailed information of the dynamic attributes associated with the variable.

The objective function (Equation (1)) minimizes the sum of penalties across all forest management goals. Penalties are aggregated from the user-defined "SOFT" non-relational constraints.

Non-relational constraints (Equations (2)–(4)): Users define parameters for each constraint including the bounds (upper limit and/or lower limit), and the per-unit magnitude of the penalty if the constraint is violated. Users also define whether a constraint is HARD, SOFT, or FREE depending on how the bounds and penalties are applied. A HARD constraint requires the value of the variable associated with the constraint to be within the specified bounds. A SOFT constraint allows for deviations from the bounds, but incurs penalties for deviations. Users can specify the penalty value so that some constraints are valued more than others. Note that penalties are applied to the per-unit deviation of the constraint and caution should be taken if mixing SOFT constraints using different units of measurement (e.g., simultaneously considering a SOFT per-volume constraint and a SOFT per-acre constraint). SOFT constraints collectively define the goal objective function where the sum of penalties across all constraints is minimized. A FREE constraint simply tracks the values of the associated variables without incurring penalties or requiring a specific level to be maintained. FREE constraints may be useful for model summarization. In PRISM, constraints may also be excluded from the problem formulation without being deleted from the GUI. This feature makes it convenient for users to find infeasibilities or conduct sensitivity analyses by quickly including or excluding in subsequent model runs.

Relational constraints (Equation (5)): The user-defined continuous variables (y_n) and their associated parameters can be considered together to form relational/flow constraints. For instance, one might define a relational constraint that states the average of timber volume harvested in periods 1–5 must be less than or equal to 20% of the harvested volume in period 6.

Transitional constraints (Equations (6)–(10)): Equation (6) describes the initial landscape condition at the beginning of the planning horizon (period 1). Equations (7) and (8) respectively capture the transition of existing forest strata areas and regenerated forest strata areas that are not being clear-cut and not subjected to lethal disturbances (LDs). In this case, age-class would increase by one when moving from a period to the next. In contrast to the above two equations, equations (9) and (10) identify strata areas that are subjected to either clear-cuts or lethal disturbances in a planning period, and require those areas to become regenerated forest strata areas at age class one in the next period. Equation (9) identifies the total area subjected to LDs in each planning period for each $s \rightarrow s'$ transition. Equation (10) uses the result from (9) plus the total area subjected to clear-cuts, then transit the whole areas to become regenerated forest strata areas at age-class one in the next period.

Illustration of building a simple model in PRISM

Here we provide an example to illustrate how PRISM can be used to evaluate a simple timber harvesting problem. For this example, we assume: 1) Users already have a forest database with sufficient information about strata and prescriptions; 2) Forest strata may be managed with any of the prescriptions associated with each stratum; 3) No disturbances will occur during the planning horizon; and 4) The cost of management activities is not considered. The model is used to schedule timber harvest through a 5-period planning horizon such that the harvest volume in every period is between 5000 and 10,000 cubic feet. There are penalties for both under harvest (2 penalty points per cubic foot lower than 5000) and over harvest (1 penalty point for each extra cubic foot greater than 10,000) which are used in the goal programming objective function to minimize the total penalty. This problem can be set up in several minutes through the two GUI windows illustrated in Fig. A2. PRISM automatically builds the complete model formulation based on information provided in the GUI, including the objective function (11), several user-defined constraints (12–15), and the core transitional constraints (6–10) as presented earlier in the Appendix.

$$Minimize: \sum_n (2l_n + u_n) \tag{11}$$

Subject to:

$$y_n = \sum_s \sum_i x_{s,i,t} + \sum_s \sum_{s'} \sum_i \overline{x}_{s,s',i,t} + \sum_s \sum_i \sum_a x'_{s,i,t,a} + \sum_s \sum_{s'} \sum_i \sum_a \overline{x}'_{s,s',i,t,a} \quad \forall n \in [1,5], t = n \tag{12}$$

$$y_n + l_n \geq 5000 \quad \forall n \in [1,5] \tag{13}$$

$$y_n - u_n \leq 10000 \quad \forall n \in [1,5] \tag{14}$$

$$y_n, l_n, u_n \geq 0 \quad \forall n \in [1,5] \tag{15}$$

Note: Harvest activities include thinning, selected cuts, clearcuts and other management activities that will contribute to harvest volume and therefore all types of decision variables can be added into the constraint (12).

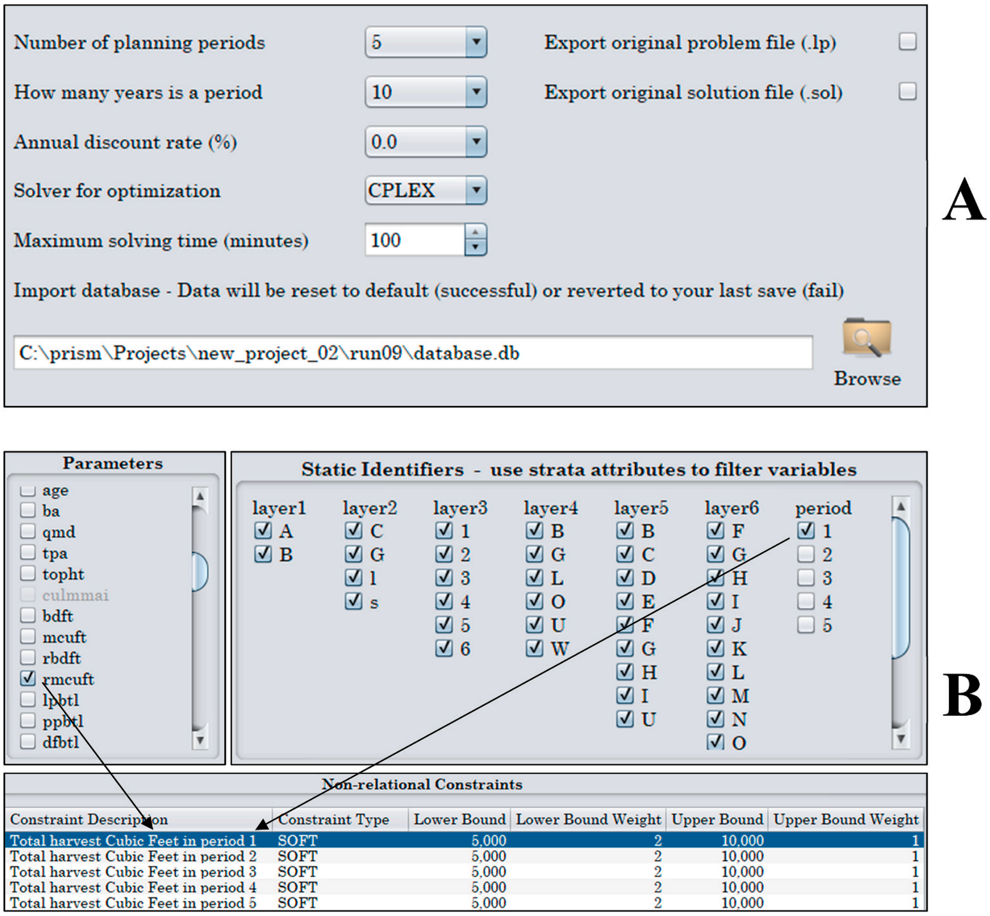

Fig. A2. The process to build an example timber harvest model in PRISM that requires using only two PRSIM GUI windows. **A:** Screenshot of a PRISM window to import database and set up some general model inputs, including the planning horizon. **B:** Another window screenshot to create user-defined constraints. Here, the two arrows represent the required selections of both "time period" and "rmcuft" to set up the constraint for timber harvest in the first planning period. The "rmcuft" parameter represents the cubic feet of harvested timber per acre; each user-defined continuous variable that represents how many acres to harvest in each period will be linked to a specific "rmcuft" value in the yield table database.

Table A1
Forestwide comparison of management alternatives in the CGNF's revised forest plan.

Issue of Emphasis	Alternative A	Alternative B	Alternative C	Alternative D	Alternative E	Alternative F
Recommended wilderness area number	7	9	9	39	0	7
Recommended wilderness area acres	33741	113382	145777	711425	0	125675
Backcountry area number	3	9	13	1	2	13
Backcountry area acres	38414	124980	299522	5937	171326	208959
Recreation emphasis area number	0	8	8	4	12	10
Recreation emphasis area acres	0	176958	160665	33408	212689	224608
Stillwater complex acres	0	101832	101832	0	101832	101832
Miles summer motorized trail no longer suitable	0	0	4	172	0	0
Miles mechanized trail no longer suitable	0	0	34	264	0	24
Acres winter motorized transport no longer suitable	0	0	24885	234431	0	10128
Forested acres suitable for timber production; percent Custer Gallatin National Forest	664,628 (22%)	573,275 (19%)	549,115 (18%)	545,274 (18%)	593,735 (19%)	565,536 (19%)
Forested acres unsuitable for timber production but where timber harvest is suitable for other purposes; percent Custer Gallatin National Forest	517,195 (17%)	595,964 (20%)	577,591 (19%)	249,141 (8%)	610,629 (20%)	614,349 (20%)
Bison	No plan direction	Proactive bison support	Proactive bison support	Most proactive bison support	Less proactive bison support	Most proactive bison support

(continued on next page)

Table A1 (*continued*)

Issue of Emphasis	Alternative A	Alternative B	Alternative C	Alternative D	Alternative E	Alternative F
Bighorn sheep disease prevention: Permitted grazing of domestic sheep or goats.	No plan direction; risk assessment per policy.	No in Pryor, AB, or MHG GAs. Yes with risk assessment elsewhere.	No in Pryor, AB, or MHG GAs. Yes with risk assessment elsewhere.	No forestwide	Yes forestwide with risk assessment	No in Pryor, AB, BBC or MHG GAs. Yes with risk assessment elsewhere.
Bighorn sheep disease prevention: Public and outfitter use of recreational pack goats	No plan direction	No in Pryor, AB, or MGH GAs. Yes with risk assessment elsewhere.	No in Pryor, AB, or MGH GAs. Yes with risk assessment elsewhere.	No forestwide	Yes forestwide with risk assessment	Yes with conditions in Pryor, AB, and MGH GAs. Yes with risk assessment elsewhere until occupied by bighorn sheep
Bighorn sheep disease prevention: Agency use of domestic sheep or goats for weed control	No plan direction; risk assessment per policy.	Yes forestwide with risk assessment	Yes forestwide with risk assessment	No forestwide	Yes forestwide with risk assessment	Yes forestwide with risk assessment
Connectivity	No plan direction	Plan components and key linkage areas	Plan components and key linkage areas	Plan components and key linkage areas	Plan components	Plan components and key linkage areas
Key linkage area acres	0	60834	59528	60834	0	62751
Aircraft landing strip acres	1022282	923303	894506	0	924574	920128

Note.
1. Reprinted from "Final Environmental Impact Statement for the 2020 Land Management Plan – Custer Gallatin National Forest", by USDA Forest Service, 2020, Summary, Table 2.
2. GA = geographic area; AB = Absaroka Beartooth Mountains Geographic Area; BBC=Bridger, Bangtail, Crazy Mountains Geographic Area; MHG = Madison, Henrys Lake, and Gallatin Mountains Geographic Area.

Table A2
Relative contributions to social and economic sustainability by alternative.

Key Social and Economic Benefit from the National Forest	Relative Contributions Greatest to Smallest (left to right)
Clean air	E, (A/B/C/D/F)
Clean water, aquatic ecosystems, and flood control	D, (B/C/F), E, *A
Conservation of wildlife and rare plants, including species for fishing, hunting, and wildlife viewing)	D, (B/C/F), E, A
Designated areas	(A/B/C/D/E/F)
Land allocations (e.g., RWA, BCA)	(B/C/D/E/F), A
Educational and volunteer programs	(B/C/D/E/F), A
Fire suppression and fuels management	D, (A/B/C/F), E
Forest products (including timber, firewood, Christmas trees, berries, mushrooms)	E, (B/C/F), D, A
Permitted livestock grazing	(A/B/C), F, (D/E)
Income (payments in lieu of taxes, secure rural schools, labor income in various industries: recreation, timber, grazing, etc.)	E, (B/C/F), D, A
Infrastructure	(A/B/C/F), E, D
Inspiration (including spiritual inspiration)	(B/C/D/E/F), A
Jobs (and induced jobs, including recreation, timber, grazing, etc.)	E, (B/C/F), D, A
Mineral and energy resources	A, E B, F, C, D
Preservation of historic, cultural, Tribal or archeological sites	D, C, F, B, A, E
Sustainable recreation	(B/C/D/E/F), A
Scenery	D, C, F, B, A, E

Note.
1. Reprinted from "Final Environmental Impact Statement for the 2020 Land Management Plan – Custer Gallatin National Forest", by USDA Forest Service, 2020, Summary, Table 9. (USDA Forest Service, 2020c).
2. Alternative A represents the current plans in this table. Alternatives are ordered left to right, from greatest to smallest contribution to social and economic sustainability. Alternatives in parentheses and separated by a slash denote similar contributions.

Table A3
Projected timber volume productions and management activities averaged for the first five decades from the six CGNF's alternatives.

PRISM's Outputs	Alt. A	Alt. B	Alt. C	Alt. D	Alt. E	Alt. F
Intermediate (acres per year)	719	672	674	609	777	668
Fuels (acres per year)	2256	2332	2330	4473	1215	2343
Regeneration (acres per year)	409	398	394	287	523	400
Prescribed fire (acres per year)	2746	2756	2766	2981	2732	2744
Total area treated (acres per year)	6130	6158	6165	8349	5247	6155
Timber sale quantity (mmbf per year)	10	10	10	6	15	10

Note.
1. Reprinted from "Final Environmental Impact Statement for the 2020 Land Management Plan – Custer Gallatin National Forest", by USDA Forest Service, 2020, Volume 3, Table 16. (USDA Forest Service, 2020c).
2. mmbf = millions of board feet.

Table A4
Sensitivity analysis based on first decade harvest volume. Each run has one set of constraints eliminated. A set of constraints is a limiting factor when eliminating them would result in a higher decade one volume.

Constraint Set	SR1	SR2	SR3	SR6	SR7	SR8	SR9	SR10
Ending Inventory	yes	yes	yes	yes	yes	yes	Yes	yes
Non-declining timber volume	yes	yes	yes	yes	yes	yes	**No**	yes
Thinning Constraints	yes	yes	yes	yes	yes	**no**	Yes	yes
Dispersion of Openings	yes	yes	yes	yes	**no**	yes	Yes	yes
Lynx Wildlife Constraints	yes	yes	yes	**no**	yes	yes	Yes	yes
Management Area Volume (80%)	yes	yes	**no**	yes	yes	yes	Yes	yes
Custer vs. Gallatin Volume (20%)	yes	**no**	yes	yes	yes	yes	Yes	yes
Prescribed Burn Constraints	**no**	yes	yes	yes	yes	yes	Yes	yes
Decade 1 mmbf volume per year	33.53	28.12	28.12	29.36	28.12	33.95	94.96	28.12

Note.
1. Reprinted from “Final Environmental Impact Statement for the 2020 Land Management Plan – Custer Gallatin National Forest”, by USDA Forest Service, 2020, Volume 3, Table 17. (USDA Forest Service, 2020c).
2. SR = sensitivity run; mmbf = millions of board feet.

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